

Let $\tilde{u}_h$ and $\tilde{R}$ be the piecewise affine interpolants of u_h^j and R^j , respectively, and put $\rho^j = R^j - u_h^j$. Write f_{lin} and ψ_{lin} for the piecewise affine interpolants of their nodal values. The backward Euler equation and the definition of ψ_h^j give $\psi_h^j = -\delta_t u_h^j$ for $j \geq 1$. Consequently, on I_j , the reconstruction equation yields the residual identity

$$\mathcal{K}(u - \tilde{R}) = f - f_{\text{lin}} - \delta_t \rho^j + \frac{t_j - t}{\tau_j} (\psi_h^j - \psi_h^{j-1}). \quad (1)$$

In particular, interpolation of the elliptic right-hand sides contributes $f - f_{\text{lin}}$; it cannot be replaced by $f - f^j$ without an additional term.

Here is an upper bound that keeps the data contribution explicit. Define its Green data-oscillation functional by

$$D_T := \left\| \int_0^T (\Gamma(T-t), f(t) - f_{\text{lin}}(t)) dt \right\|_{\infty, \Omega}, \quad (2)$$

where the convolution is understood by transposition if its displayed pointwise integral does not exist. Set $\phi_0(s) = \kappa_0 e^{-\gamma s}$, and let $\eta_j = \eta(u_h^j, f^j + \psi_h^j)$ and $\eta_{\delta,j} = \eta(\delta_t u_h^j, \delta_t (f^j + \psi_h^j))$. Then

$$\begin{aligned} \|u(T) - u_h^M\|_{\infty, \Omega} &\leq \eta_M + \phi_0(T) (\|u_0 - u_h^0\|_{\infty, \Omega} + \eta_0) + D_T \\ &\quad + \sum_{j=1}^M \int_{I_j} \phi_0(T-t) \left(\eta_{\delta,j} + \frac{t_j - t}{\tau_j} \|\psi_h^j - \psi_h^{j-1}\|_{\infty, \Omega} \right) dt. \end{aligned}$$

Indeed, time independence of $\mathcal{M}$ makes $\delta_t R^j$ the elliptic reconstruction of $\delta_t u_h^j$. Assumption 2 therefore bounds $\|\rho^j\|_{\infty}$ by η_j and $\|\delta_t \rho^j\|_{\infty}$ by $\eta_{\delta,j}$. Applying the Green representation to (1), then using the L^1 Green bound on every term other than the data convolution, gives (3), since $u(T) - u_h^M = (u(T) - R^M) + \rho^M$. For the weak solution u , this use of the representation is understood by transposition: apply the representation lemma to temporally regularised problems and pass to the weak-solution limit. One cannot apply that lemma directly to $u - \tilde{R}$, because the stated assumptions do not give its time derivative in $L^2(0, T; H)$.

The distinction concerning D_T is essential. If $f - f_{\text{lin}} \in L^1(0, T; L^\infty(\Omega))$, Assumption 1 supplies the directly bounded data indicator

$$D_T \leq \int_0^T \phi_0(T-t) \|f(t) - f_{\text{lin}}(t)\|_{\infty, \Omega} dt. \quad (4)$$

The stated hypothesis $f \in L^2(0, T; H_0^1(\Omega))$ does **not** imply that the right-hand side of (4) is finite. Thus (3) is a bound with a Green data-oscillation

functional, but the supplied assumptions do not establish a finite upper bound for that functional using only the stated Green constants and forcing norms. A finite, fully estimator-based version of (3) requires an additional assumption or an independently certified bound for D_T ; omitting that qualification would not prove the requested finite a posteriori estimate.